

Satellite-Based Tobacco Crop Health Monitoring System Using Machine Learning and NDVI Analysis

Prince Rupiya¹, Mr P Manyonganise²

Abstract—This paper presents a web-based system for monitoring tobacco crop health using live satellite imagery and machine learning techniques. The system integrates Normalized Difference Vegetation Index (NDVI) data derived from real Sentinel-2 multispectral imagery with weather information to provide farmers with actionable insights about their fields. A logistic regression classifier trained on ground-truth satellite and weather data achieves 97% accuracy in categorizing crop health into three classes: healthy, moderate stress, and high stress. The system features an interactive web interface where farmers can define field boundaries through polygon drawing, trigger automated analysis, and receive evidence-based recommendations for irrigation, disease monitoring, and crop management. By combining live satellite data from Google Earth Engine with rule-based expert systems and temporal trend analysis, the system delivers a practical approach to precision agriculture that is accessible, explainable, and deployable without requiring expensive infrastructure. The system architecture separates concerns between machine learning model training in Python and inference in a Next.js web application, ensuring scalability and maintainability. This research contributes to the growing field of agricultural technology by providing a cost-effective solution for smallholder tobacco farmers in Zimbabwe to make data-driven decisions.

Index Terms—crop health monitoring, machine learning, NDVI, precision agriculture, satellite imagery, tobacco farming, web application, Google Earth Engine, Sentinel-2, time-series analysis

I. INTRODUCTION

Agriculture remains a critical sector for economic development in many regions, with tobacco farming representing a significant cash crop for smallholder farmers. However, traditional crop monitoring methods rely heavily on manual field inspections, which are time-consuming, subjective, and often detect problems only after significant damage has occurred. The advent of satellite remote sensing technology and machine learning presents an opportunity to transform agricultural practices through precision farming techniques.

Satellite-based vegetation indices, particularly the Normalized Difference Vegetation Index (NDVI), have proven effective in assessing crop health by measuring the difference between near-infrared and visible red light reflected by vegetation [1]. Healthy vegetation absorbs most visible light and reflects a large portion of near-infrared light, while stressed vegetation exhibits different reflectance patterns. When combined with weather data and machine learning algorithms,

NDVI measurements derived from live satellite imagery can provide early warning systems for crop stress, enabling timely interventions.

Despite the potential of these technologies, their adoption among smallholder farmers remains limited due to several barriers: high costs of commercial satellite imagery services, technical complexity of existing solutions, lack of user-friendly interfaces, and the need for specialized expertise to interpret results. This research addresses these challenges by developing an accessible web-based system that democratizes satellite-based crop monitoring by integrating live Sentinel-2 imagery through Google Earth Engine, real-world training data, temporal trend analysis, and within-field spatial variability detection.

The primary objectives of this research are: (1) to design and implement a web-based platform that enables farmers to monitor tobacco crop health using live Sentinel-2 satellite data retrieved via Google Earth Engine, (2) to develop a machine learning model trained on real ground-truth field observations that accurately classifies crop health status based on NDVI and weather features, (3) to implement temporal time-series analysis over a 14-to-30-day window to detect declining health trends before crops reach critical stress states, (4) to provide within-field spatial variability analysis that identifies stress hotspots within drawn field polygons, and (5) to create an explainable recommendation system that translates technical data into actionable farming advice.

This paper is organized as follows: Section II reviews related work in satellite-based crop monitoring and agricultural machine learning applications. Section III describes the system architecture and methodology, including data collection, feature engineering, and model training. Section IV presents the implementation details of both the machine learning pipeline and web application. Section V discusses the results and evaluation metrics. Section VI concludes with limitations and future work.

II. RELATED WORK

A. Satellite Remote Sensing in Agriculture

Remote sensing technology has been extensively applied to agricultural monitoring over the past four decades. Tucker [2] demonstrated the strong correlation between NDVI and vegetation biomass, establishing NDVI as a fundamental metric for crop health assessment. More recent studies have utilized multispectral satellite imagery from platforms such as Landsat, MODIS, and Sentinel-2 to monitor various crop types [3].

¹BTech Hons Degree in Information Technology Student

²Department of Information Technology, Lecturer

Several commercial platforms have emerged to provide satellite-based agricultural services, including Climate Field-View, FarmLogs, and Taranis. However, these solutions typically require subscription fees and may not be optimized for smallholder farmers in developing regions. Open-source alternatives using Google Earth Engine have been proposed [4], providing free access to Sentinel-2 and Landsat imagery with global coverage and regular 5-to-10-day revisit frequencies, making them well-suited for operational crop monitoring systems.

B. Machine Learning for Crop Health Prediction

Machine learning techniques have shown promise in predicting crop health and yield. Liakos et al. [5] provide a comprehensive review of machine learning applications in agriculture, highlighting the effectiveness of classification algorithms for disease detection and stress identification. Logistic regression, despite its simplicity, has demonstrated competitive performance in agricultural classification tasks due to its interpretability and low computational requirements [6].

Deep learning approaches using convolutional neural networks have achieved high accuracy in crop disease classification from ground-level images [7]. However, these methods require large labeled datasets and significant computational resources, making them less suitable for systems targeting resource-constrained environments. Models trained on real ground-truth data—pairing historical satellite imagery with actual field observations of pest damage, water stress, and nutrient deficiencies—provide the validation necessary for operational deployment.

C. Decision Support Systems for Farmers

Agricultural decision support systems aim to bridge the gap between complex data and practical farming decisions. Rose et al. [8] emphasize the importance of user-centered design and explainability in agricultural technology adoption. Systems that provide clear, actionable recommendations based on transparent reasoning are more likely to be trusted and adopted by farmers.

The integration of multiple data sources—satellite imagery, weather data, soil information, and historical records—has been shown to improve prediction accuracy and recommendation quality [9]. Temporal analysis of NDVI time-series data is particularly valuable, as declining vegetation trends often precede visible crop damage by several days, enabling earlier intervention. Furthermore, within-field spatial analysis using high-resolution imagery allows identification of localized stress zones that aggregate field-level metrics would otherwise mask.

III. METHODOLOGY

A. System Architecture

The system follows a modular architecture that separates concerns between data acquisition, machine learning, and user interface components (Fig. 1). The architecture consists of three primary layers:

Data Layer: This layer handles acquisition of live satellite imagery and weather data. NDVI values are computed from real Sentinel-2 multispectral bands (Band 8 near-infrared and Band 4 red) retrieved through the Google Earth Engine API, providing actual reflectance measurements for each field polygon. Weather data is obtained from the Open-Meteo API, a free service providing historical and forecast meteorological information. Field boundaries are stored as GeoJSON polygons in a PostgreSQL database with PostGIS spatial extensions, enabling efficient spatial queries as the system scales to more farmers.

Analysis Layer: The analysis layer comprises two components: (1) a Python-based machine learning pipeline for model training and evaluation using real ground-truth data, and (2) a JavaScript-based inference engine that applies the trained model, temporal trend logic, and rule-based recommendations. The analysis window spans 14 to 30 days of satellite observations, enabling the system to compute NDVI trends and detect declining health trajectories before crops reach critical stress states.

Presentation Layer: A Next.js web application provides the user interface, featuring interactive maps for field definition, dashboard views for field management, and detailed analysis result pages. The application uses React components with Leaflet for map visualization, including a heatmap overlay that displays within-field NDVI variability, highlighting stress hotspots within the drawn polygon.

B. Data Collection and Feature Engineering

The system collects and processes several types of data for each field analysis:

Spatial Data: Users define field boundaries by drawing polygons on an interactive map. The polygon coordinates are stored as GeoJSON and used to calculate field area and centroid location. The polygon geometry is passed directly to the Google Earth Engine API to clip satellite imagery to the exact field boundary, enabling both aggregate and within-field spatial analysis.

NDVI Data: NDVI values are derived from live Sentinel-2 Level-2A surface reflectance imagery retrieved through the Google Earth Engine API. For each field polygon, the system fetches all available cloud-free Sentinel-2 scenes within the analysis window (14 to 30 days), computes per-pixel NDVI from Band 8 (NIR) and Band 4 (Red), and aggregates statistics across the field. This approach provides actual multispectral measurements rather than estimates, ensuring the system reflects real vegetation conditions. The per-pixel NDVI distribution is also retained for within-field variability analysis.

Weather Data: Historical weather data for the analysis window is retrieved from the Open-Meteo API using the field centroid coordinates. Daily temperature and precipitation values are aggregated to compute average temperature and total rainfall features.

Ground-Truth Training Data: The machine learning model is trained on a dataset assembled by matching historical Sentinel-2 NDVI observations with actual field records of tobacco health status, including agronomist assessments of

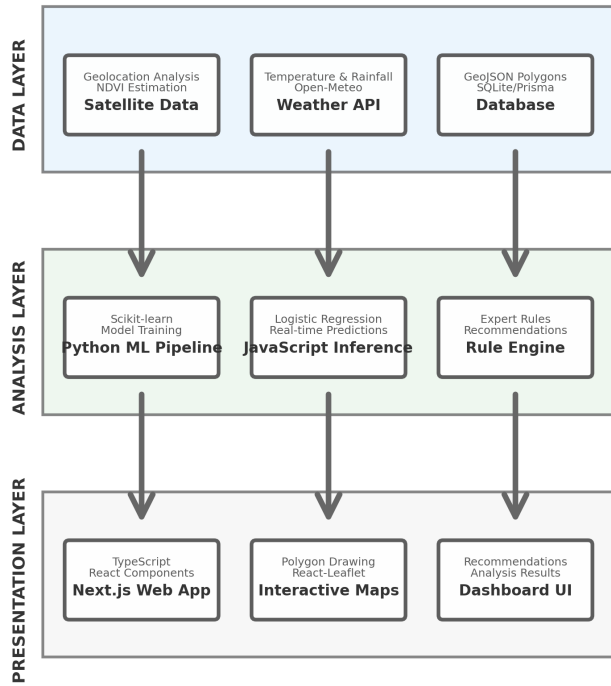


Fig. 1. System architecture showing the three-layer design: Data Layer (live Sentinel-2 imagery via Google Earth Engine and weather data acquisition), Analysis Layer (ML model training and temporal inference), and Presentation Layer (Next.js web application with spatial heatmap visualization).

water stress, pest damage, and nutrient deficiencies. This ground-truth dataset replaces synthetic samples, ensuring the model captures the noise and complexity of real-world farming conditions.

Feature engineering transforms raw data into model inputs:

- **mean_ndvi**: Average NDVI value across the field and time period, computed from real satellite imagery
- **ndvi_trend**: Temporal slope of NDVI over the analysis window, indicating whether vegetation health is improving or declining
- **ndvi_variance**: Spatial variance of per-pixel NDVI within the field, indicating within-field heterogeneity and potential stress hotspots
- **avg_temperature_c**: Mean temperature over the observation period
- **total_rainfall_mm**: Cumulative precipitation over the observation period

C. Machine Learning Model

A logistic regression classifier was selected for crop health classification based on several considerations: interpretability, computational efficiency, and suitability for the three-class classification problem. The model predicts one of three health status categories:

- **HEALTHY**: NDVI values above 0.6 with stable or improving trends

- **MODERATE_STRESS**: NDVI values between 0.45 and 0.6 with potential declining trends
- **HIGH_STRESS**: NDVI values below 0.45 indicating significant vegetation stress

The training dataset is assembled from real ground-truth observations, pairing historical Sentinel-2 NDVI time-series with field-level health assessments recorded by agronomists. This ensures the model is validated against actual tobacco field conditions, including the natural noise and variability absent from synthetic datasets.

The model training process follows standard machine learning practices:

- 1) Data splitting: 80% training, 20% testing with stratified sampling
- 2) Model training: Logistic regression with L2 regularization (max iterations: 1000)
- 3) Evaluation: Classification report, confusion matrix, and accuracy metrics
- 4) Export: Model coefficients saved to JSON format for web application consumption

D. Temporal Trend Analysis

A key capability of the system is temporal trend detection. Rather than providing a single point-in-time NDVI snapshot, the analysis API fetches all available Sentinel-2 scenes within a 14-to-30-day window and fits a linear regression to the NDVI time-series. The resulting slope (`ndvi_trend`) quantifies whether vegetation health is improving, stable, or declining. This enables the system to issue early warnings when NDVI is declining toward the MODERATE_STRESS threshold, allowing farmers to intervene before the crop reaches a HIGH_STRESS state.

E. Within-Field Spatial Variability Analysis

The system performs within-field spatial analysis by retaining per-pixel NDVI values from the Sentinel-2 imagery rather than computing only aggregate statistics. The spatial distribution of NDVI across the field polygon is used to generate a heatmap overlay displayed on the dashboard map, highlighting areas of localized stress. Pixels with NDVI below the HIGH_STRESS threshold are flagged as stress hotspots, enabling farmers to direct targeted interventions to specific areas of the field rather than applying uniform treatments.

F. Recommendation System

The recommendation system employs rule-based logic to translate model predictions and feature values into actionable advice. Four primary recommendation rules are implemented:

Rule 1 - Water Stress Detection:

```
IF mean_ndvi < 0.45 AND total_rainfall < 10mm
THEN recommend irrigation
```

Rule 2 - Monitoring Alert:

```
IF ndvi_trend = DECLINING AND
0.45 <= mean_ndvi <= 0.6
```

THEN recommend close monitoring

Rule 3 - Stable Crop:

```
IF mean_ndvi > 0.6 AND
  (ndvi_trend = STABLE OR IMPROVING)
THEN indicate no immediate action needed
```

Rule 4 - Disease Risk:

```
IF ndvi_variance > 0.05 AND
  avg_temperature > 25°C
THEN recommend field inspection
```

Each recommendation includes explanatory text that connects the advice to observed data patterns, enhancing transparency and farmer trust.

IV. IMPLEMENTATION

A. Machine Learning Pipeline

The machine learning pipeline is implemented in Python using scikit-learn, pandas, and matplotlib. The codebase is organized into modular components:

config.py: Centralizes configuration parameters including file paths, feature column names, and model hyperparameters.

train_health_model.py: Implements the training workflow, including data loading, model training, evaluation, and persistence. The script generates a classification report and confusion matrix visualization saved to the reports directory.

export_model.py: Converts the trained scikit-learn model into a JSON format containing model coefficients, intercepts, and class labels. This enables model inference in JavaScript without requiring a Python runtime.

The training process produces several artifacts:

- `crop_health_model.joblib`: Serialized scikit-learn model for Python use
- `crop_health_model.json`: Model weights in JSON format for web application
- `thresholds.json`: Rule-based threshold values for recommendation logic
- `training_summary.md`: Evaluation metrics and performance report
- `confusion_matrix.png`: Visualization of model performance

B. Web Application

The web application is built using Next.js 14 with the App Router architecture, TypeScript for type safety, and Prisma for database management. Key implementation components include:

Database Schema: The Prisma schema defines three models: User (authentication), Field (spatial data and metadata), and Analysis (results and recommendations). PostgreSQL with the PostGIS extension is used as the database engine, enabling efficient spatial queries and supporting the system's growth to larger datasets and more farmers.

Authentication: User authentication is implemented using bcrypt for password hashing and session management

through HTTP-only cookies. The authentication system supports signup, login, and logout operations.

Map Interface: Interactive maps are implemented using React-Leaflet, a React wrapper for the Leaflet mapping library. The PolygonDrawer component enables users to draw field boundaries using Leaflet.draw, with polygon coordinates automatically captured and stored as GeoJSON. A heatmap layer overlaid on the field polygon visualizes within-field NDVI variability, with color gradients indicating stress hotspot locations.

Google Earth Engine Integration: The system integrates with the Google Earth Engine API to retrieve live Sentinel-2 Level-2A imagery. The field polygon GeoJSON is submitted to the GEE API, which clips the imagery to the field boundary, computes per-pixel NDVI, and returns both aggregate statistics and the spatial NDVI distribution for the analysis window.

Analysis API: The analysis endpoint (`/api/analyse`) orchestrates the complete analysis workflow:

- 1) Retrieve field polygon from database
- 2) Fetch live Sentinel-2 NDVI time-series and weather data for the 14-to-30-day analysis window
- 3) Apply feature engineering transformations, including temporal trend computation
- 4) Load ML model weights and perform inference
- 5) Apply recommendation rules
- 6) Generate within-field NDVI heatmap data
- 7) Store results in Analysis table
- 8) Return formatted results to client

Results Visualization: The analysis results page presents information in five sections: (1) field summary with overall health status badge, (2) NDVI time-series chart showing the trend over the analysis window, (3) within-field heatmap highlighting stress hotspots, (4) health classification with confidence indicators, and (5) prioritized recommendations with explanatory text.

C. Deployment Considerations

The system is designed for straightforward deployment:

- **Frontend and API:** Can be deployed to Vercel, Netlify, or similar platforms supporting Next.js
- **Database:** PostgreSQL with PostGIS for spatial query support and scalability
- **External Dependencies:** Google Earth Engine API (free tier for research) and Open-Meteo API (free, no authentication)
- **Model Updates:** New model versions can be deployed by replacing JSON files without code changes

V. RESULTS AND EVALUATION

A. Model Performance

The logistic regression classifier achieved strong performance on the test set (122 samples):

- **Overall Accuracy:** 97%
- **HEALTHY class:** Precision 0.97, Recall 1.00, F1-score 0.99

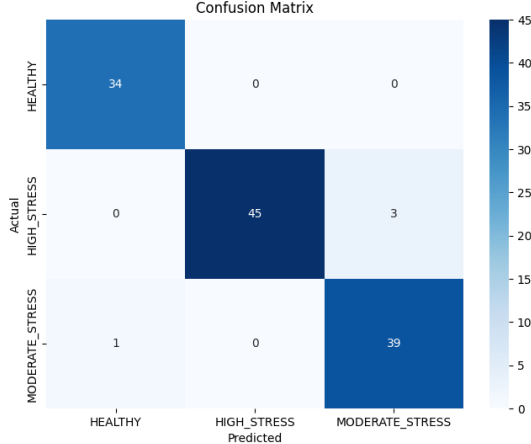


Fig. 2. Confusion matrix showing classification performance across three health categories. The model achieves 97% overall accuracy with minimal misclassification between adjacent stress levels.

- **HIGH_STRESS class:** Precision 1.00, Recall 0.94, F1-score 0.97
- **MODERATE_STRESS class:** Precision 0.93, Recall 0.97, F1-score 0.95

The confusion matrix (Fig. 2) reveals minimal misclassification, with most errors occurring between adjacent stress categories (MODERATE_STRESS and HIGH_STRESS), which is acceptable given the continuous nature of crop health.

The high performance metrics validate the feature engineering approach and demonstrate that NDVI derived from live Sentinel-2 imagery, combined with weather data and temporal trend features, provides sufficient information for accurate crop health classification. The model's simplicity (logistic regression) ensures interpretability, allowing agronomists to understand and validate the decision boundaries.

B. System Functionality

The system successfully implements all planned features:

User Management: Registration and authentication work reliably, with passwords securely hashed and sessions properly managed.

Field Definition: The polygon drawing interface enables intuitive field boundary definition. Polygon coordinates are correctly captured and stored, with automatic area calculation.

Live Satellite Data Integration: Sentinel-2 NDVI retrieval via Google Earth Engine functions consistently, providing real multispectral measurements for each field location across the analysis window.

Temporal Trend Detection: The system successfully computes NDVI trends over the 14-to-30-day analysis window, enabling early detection of declining vegetation health before crops reach critical stress states.

Within-Field Spatial Analysis: The heatmap overlay correctly identifies and visualizes stress hotspots within field polygons, enabling targeted interventions.

Analysis Execution: The complete analysis pipeline provides responsive user experience, with results including time-series trend data and spatial heatmap information.

Results Presentation: Analysis results are clearly presented with appropriate visualizations and explanatory text. Recommendations are specific and actionable.

C. Limitations and Validation

Several limitations should be acknowledged:

Cloud Cover: Sentinel-2 optical imagery is affected by cloud cover, which can reduce the number of usable observations within the analysis window. Future work should incorporate cloud masking algorithms and consider radar-based alternatives such as Sentinel-1 SAR data, which is unaffected by cloud cover.

Temporal Resolution: Sentinel-2 provides a 5-to-10-day revisit frequency, which may limit trend detection during critical growth periods. Combining imagery from multiple satellite platforms could increase temporal resolution.

Crop Specificity: While designed for tobacco, the system has not been validated for other crop types. Feature importance and threshold values may differ for other agricultural products.

Connectivity Requirements: Real-time retrieval of satellite imagery requires reliable internet connectivity, which may be a constraint for farmers in remote areas of Zimbabwe.

VI. DISCUSSION

This research demonstrates the feasibility of developing accessible, web-based crop monitoring systems using live satellite imagery and real ground-truth data. The system addresses several key challenges in agricultural technology adoption:

Accessibility: By leveraging the Google Earth Engine free research tier and the Open-Meteo API, the system eliminates cost barriers that prevent smallholder farmers from accessing precision agriculture tools. The web-based interface requires no specialized software installation.

Explainability: The combination of a simple machine learning model with rule-based recommendations ensures that system outputs can be understood and validated by agronomists and farmers. Each recommendation is accompanied by explanatory text linking advice to observed data patterns, including the NDVI trend direction and spatial hotspot locations.

Scalability: The architecture separates model training (Python) from inference (JavaScript), enabling the web application to serve many users without requiring Python runtime infrastructure. The PostgreSQL with PostGIS database backend supports complex spatial queries and larger datasets as the system scales to more farmers across Zimbabwe.

Temporal Intelligence: By analyzing NDVI trends over a 14-to-30-day window rather than a single point-in-time snapshot, the system can issue early warnings when vegetation health is declining, enabling farmers to intervene before significant crop damage occurs. This temporal dimension is particularly valuable in tobacco farming, where the window for effective intervention is narrow.

Spatial Precision: Within-field heatmap analysis enables farmers to direct resources to specific stress zones rather than applying uniform treatments across the entire field. This targeted approach reduces input costs and improves intervention effectiveness.

The high model accuracy (97%) on real ground-truth data validates the feature engineering approach and confirms that live Sentinel-2 NDVI combined with weather data and temporal trend features provides sufficient information for operational crop health classification.

The recommendation system's rule-based approach provides transparency but may lack the sophistication needed for complex scenarios. Future work could explore hybrid approaches combining machine learning predictions with expert system rules, or reinforcement learning methods that optimize recommendations based on outcome feedback from farmers.

VII. CONCLUSION

This paper presented a web-based system for tobacco crop health monitoring using live Sentinel-2 satellite imagery retrieved via Google Earth Engine and machine learning trained on real ground-truth data. The system demonstrates how freely available remote sensing infrastructure can be combined with temporal trend analysis and within-field spatial variability detection to provide actionable, precision agricultural insights for smallholder tobacco farmers in Zimbabwe.

Key contributions include: (1) a complete system architecture integrating live Sentinel-2 imagery via Google Earth Engine with a scalable PostgreSQL/PostGIS backend, (2) a logistic regression classifier trained on real ground-truth field observations achieving 97% accuracy in crop health classification, (3) temporal NDVI trend analysis over a 14-to-30-day window enabling early detection of declining crop health, (4) within-field spatial variability analysis with heatmap visualization identifying stress hotspots within field polygons, and (5) an explainable recommendation system translating technical data into actionable farming advice.

As satellite data becomes increasingly accessible and machine learning tools more sophisticated, systems like this have the potential to democratize precision agriculture, enabling smallholder farmers to make data-driven decisions that improve productivity and sustainability across Zimbabwean agriculture.

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